

12 mJ per Class On-Device Online Few-Shot Class-Incremental Learning

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Abstract—Few-Shot Class-Incremental Learning (FSCIL) enables machine learning systems to expand their inference capabilities to new classes using only a few labeled examples, without forgetting the previously learned classes. Classical backpropagation-based learning and its variants are often unsuitable for battery-powered, memory-constrained systems at the extreme edge. In this work, we introduce Online Few-Shot Class-Incremental Learning (O-FSCIL), based on a lightweight model consisting of a pre-trained and metalearned feature extractor and an expandable explicit memory storing the class prototypes. The architecture is pretrained with a novel feature orthogonality regularization and metalearned with a multi-margin loss. For learning a new class, our approach extends the explicit memory with novel class prototypes, while the remaining architecture is kept frozen. This allows learning previously unseen classes based on only a few examples with one single pass (hence online). O-FSCIL obtains an average accuracy of 68.62% on the FSCIL CIFAR100 benchmark, achieving state-of-the-art results. Tailored for ultra-low-power platforms, we implement O-FSCIL on the 60mW GAP9 microcontroller, demonstrating online learning capabilities within just 12mJ per new class.

Index Terms—Continual learning, on-device learning, deep neural networks, microcontrollers.

I. INTRODUCTION

Classical Machine Learning (ML) solutions often use large datasets to train a highly complex yet fixed model, which cannot adapt to the needs and requirements of the end user. Such systems are nonetheless exposed to dynamic, ever-changing environments; thus, adaptability is a crucial requirement for an intelligent system. Moreover, while in a server-based, offline learning paradigm, curated and labeled data is widely available, that is seldom the case when a pretrained model must adapt to a particular user. FSCIL evaluates models on two aforementioned problems, requiring a previously trained model to expand its classification domain while providing very few labeled training samples. The key challenge is to prevent catastrophic forgetting (i.e., forgetting prior knowledge during knowledge expansion) and avoid overfitting to the few novel samples. These goals can be achieved by balancing stability and plasticity [1]. Several FSCIL solutions have tackled sample-scarce class learning over several incremental sessions [2], [3] in a transfer learning fashion, where a model is pretrained to classify a set of predefined base classes, followed by freezing its backbone and only training its classification head to learn novel classes.

Although such methods achieve remarkable results on previously unseen classes, they rely on the computationally expensive backpropagation algorithm. Alternative solutions, such as Neural Collapse Few-Shot Class-Incremental Learning (NC-FSCIL) [4], Semantic-Aware Virtual Contrastive (SAVC) model [5], or Constrained Few-Shot Class-Incremental Learning (C-FSCIL) [6], [7], use large ResNet backbones [8]. However, adapting to new requirements and learning novel classes should happen directly on the user device, thus addressing privacy and security concerns [9], while also avoiding energy-hungry data streaming to the cloud. Therefore, when employed at the extreme edge, FSCIL algorithms should operate under the TinyML constraints [10]; hence, suitably small and computationally affordable backbones are required.

On-device training frameworks targeting Microcontroller Units (MCUs) have been proposed [11]–[13], which encouraged the development of domain adaptation [14] and class continual learning [15] strategies tailored for MCUs. However, such iterative learning approaches require storing training samples during the update process. Our proposed online training methodology addresses the storage constraints by enabling novel class learning with a single pass over the available samples. This also decreases the learning time and, thus, the energy consumption of our system. Moreover, as opposed to FSCIL strategies deployed on Tensor Processing Units (TPUs) [16], Field Programmable Gate Arrays (FPGAs) [17], neuromorphic chips [18], or in-memory processors [7], our methodology is suitable for off-the-shelf, widely available MCUs, with a conservative power profile.

This work introduces O-FSCIL¹, a lightweight FSCIL learning methodology aimed at resource-constrained edge devices. Using orthogonal regularization and data augmentation strategies such as Mixup [19] and Cutmix [20], we pretrain the MobileNetV2 [21] backbone, which requires only 2.5 million parameters and 149.2 million Multiply-Accumulate (MAC) operations. During the server-side metalearning phase, we employ a multi-margin loss to prevent overfitting to the few labeled samples. We further demonstrate the on-device learning capabilities by deploying and evaluating O-FSCIL on ultra-low-power, commercially available GAP9 [22].

The main contributions of the paper are as follows:

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¹The code will be open-sourced at: <https://github.com/pulp-platform/fscil>